Developmental science innovating Explainable AI

Katharina J. Rohlfing
Developmental Psycholinguistics

PADERBORN UNIVERSITY



CLASP research seminar
University of Gothenburg

Dear XAI Community, We Need to Talk!

Fundamental Misconceptions in Current XAI Research

Timo Freiesleben[1](✉) and Gunnar König[2]

[1] Cluster of Excellence "Machine Learning in Science", Tübingen University, Tübingen, Germany

KI - Künstliche Intelligenz
https://doi.org/10.1007/s13218-020-00637-y

DISCUSSION

One Explanation Does Not Fit All

The Promise of Interactive Explanations for Machine Learning Transparency

Kacper Sokol[1] · Peter Flach[1]

XAI: where are we standing?

Explainable AI is Dead, Long Live Explainable AI!

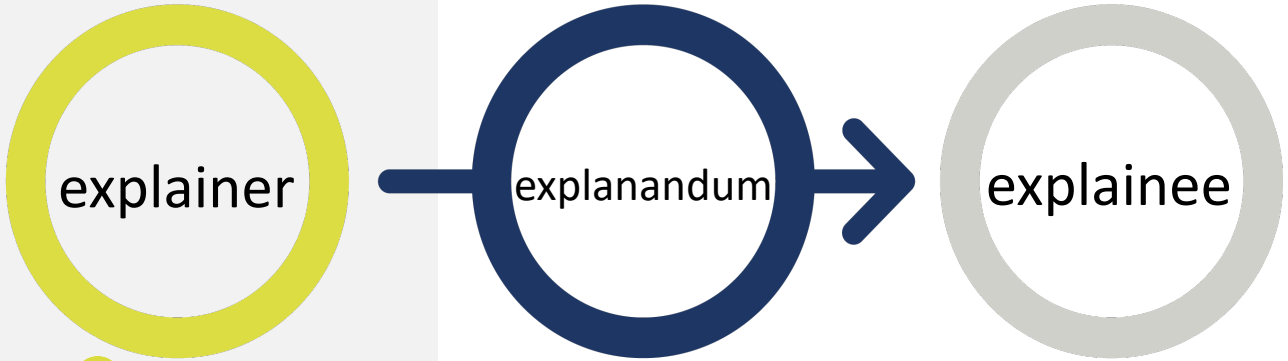
Hypothesis-driven Decision Support using Evaluative AI

Tim Miller
tmiller@unimelb.edu.au
The University of Melbourne
Melbourne, VIC, Australia

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XAI and interaction

- One way explanation
- Best fit to the explanation needs
- Explanation as a transfer of (verbal) information (Chromick & Butz, 2011)



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Developmental science innovates XAI research by providing crucial insights into:

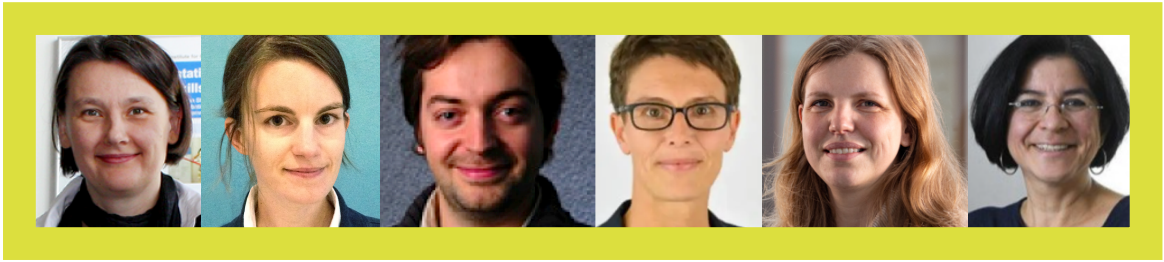
- Patternedness
- Incrementality
- Multimodality

Interactivity & Dynamicity



Developmental science innovates XAI research by providing crucial insights into:

- **Patternedness**
- Incrementality
- Multimodality



Britta Wrede, Anna-Lisa Vollmer, Pierre-Yves Oudeyer, Vivien Heller, Angela Grimminger, Juana Salas Poblete

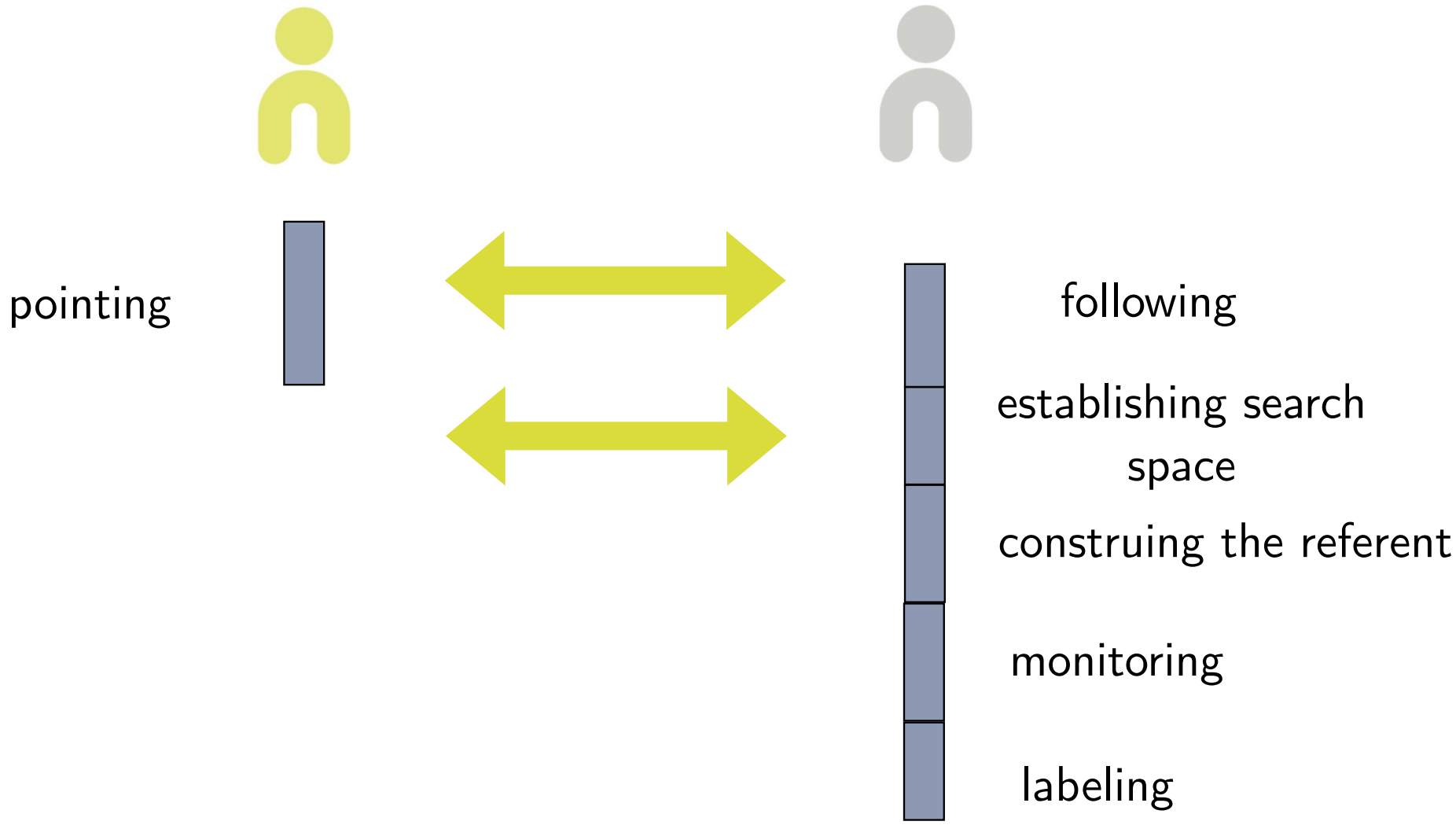
Patternedness

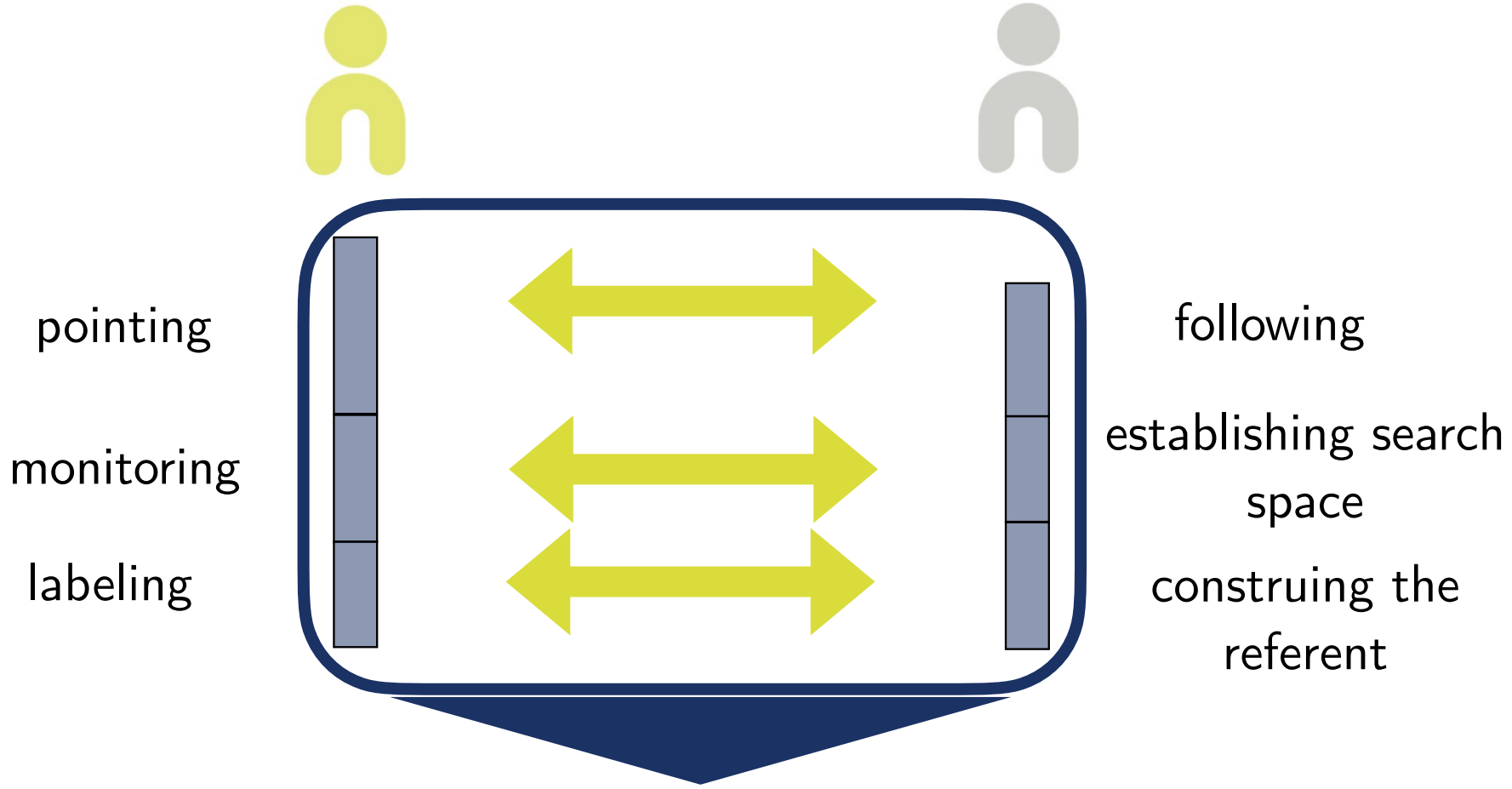
- Interaction is patterned (e.g., Bruner, 1983)
- A pattern is
  - a sequence of overt and covert actions
  - task specific
  - dyadic

video

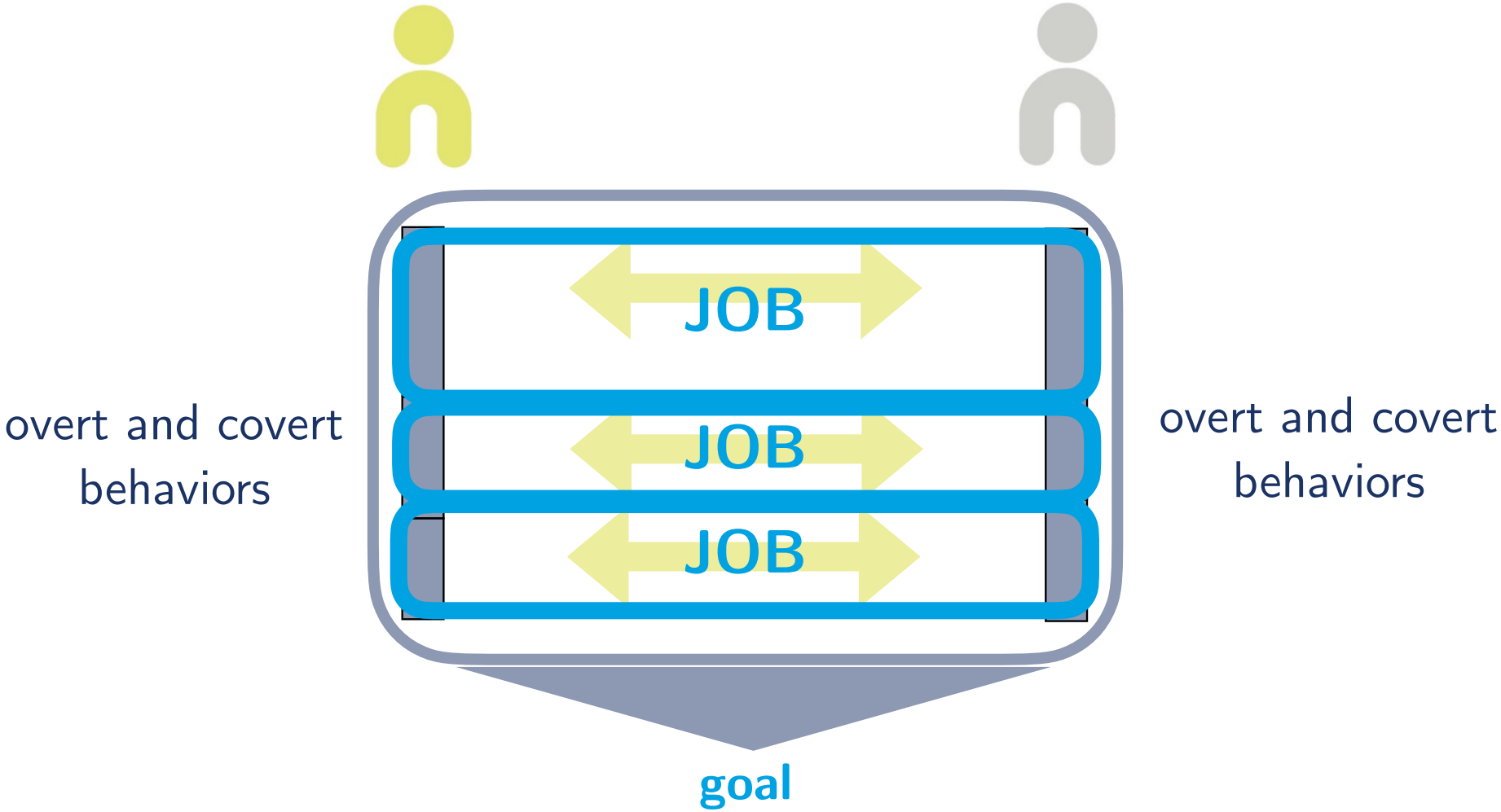


Grimminger et al., 2016; Lüke et al., 2017


pointing
following
establishing search space
construing the referent
monitoring
labeling



Rohlfing et al., 2016; Heller & Rohlfing, 2017; Rohlfing et al., 2019



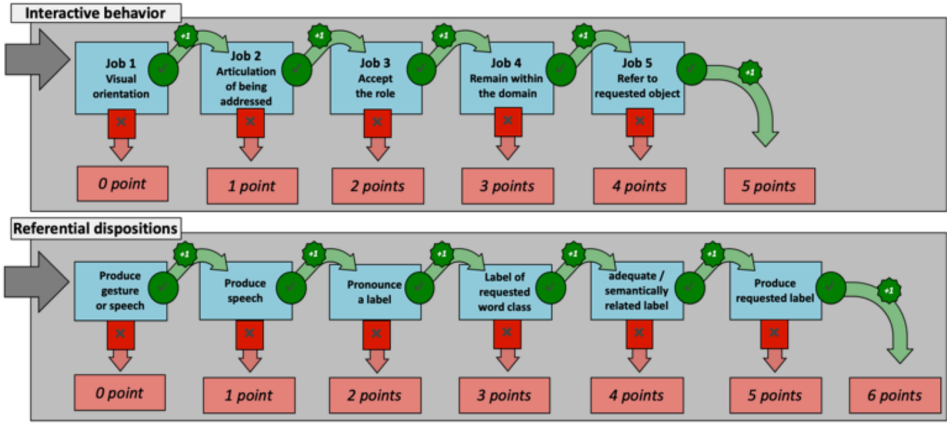
Pragmatic Frames: Rohlfing et al., 2016; Heller & Rohlfing, 2017; Rohlfing et al., 2019

video



Utilizing Pragmatic Frames

- Bottom-up approach: looking at data and analyzing the individual variation
- Identifying a schema
  - interactive behaviors
  - cognitive behaviors
- Comparing 27- with 21-month-olds, we found that
  - increasingly, children are able to master the recall of a label
  - surprisingly, more 27-month-olds did not attempt to fulfill the task (task awareness)

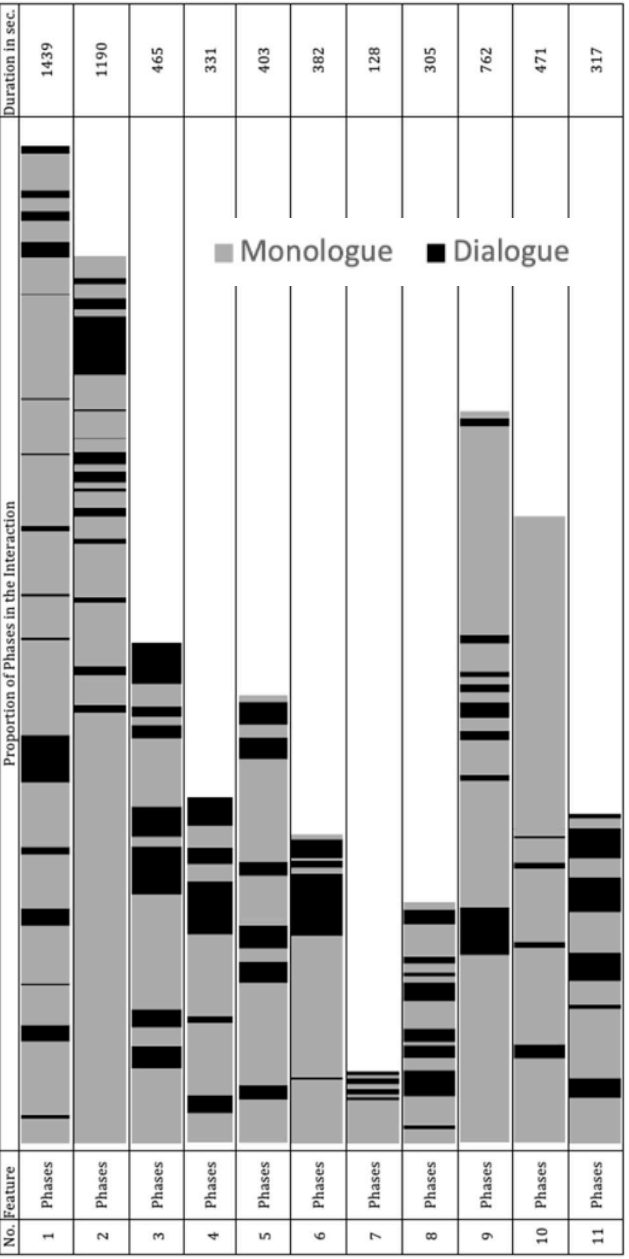


Transfer to studies on explaining

1	2	3	4	5
Establishing topical relevance	Constituting explanandum	Explicating relationships	Closing	Transferring (to another task)

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Morek et al., 2017; Lohmar, 2026



Transfer to studies on explaining

- Explanations follow a structure with a goal but the jobs are recurring (Lohmar, 2026)
- The goal of an explanation is to understand, but there exist different forms (Buschmeier et al., 2025)
- Both partners contribute to the explanations (Fisher, 2025) even though explicating relationships was executed within monological phases to 83 % (Fisher et al., 2023)
- Jobs take place within monological and dialogical phases (Fisher et al., 2023)

Some open questions

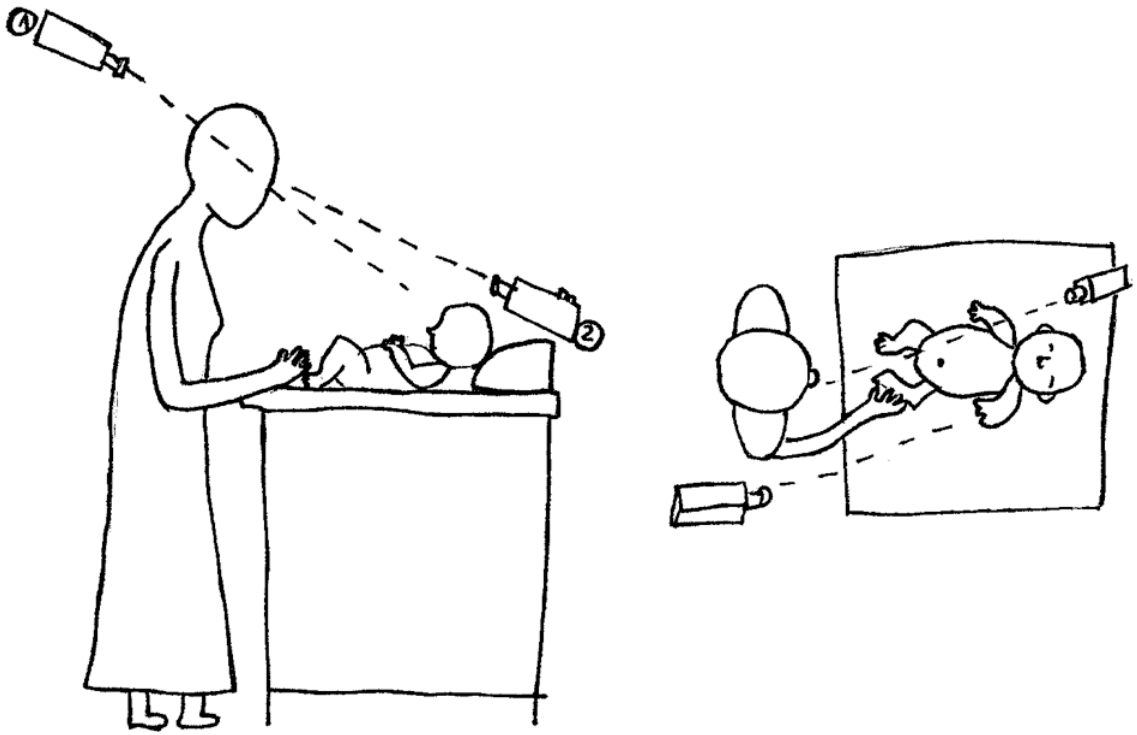
- How can the structure function as ‚local contexts' and influence dialogical behaviors?
- Why isn't an explanation following a simple structure? In other words: Why are jobs re-done? What are the cognitive functions of the local contexts?

Developmental science innovates XAI research by providing crucial insights into:

- Patternedness
- **Incrementality**
- Multimodality



Yukie Nagai, Iris Nomikou, Joanna Rączaszek-Leonardi, Giuseppe Leonardi, Karola Pitsch



- Video recordings of everyday interactions in Germany ($N = 17$)
- 6 monthly visits to families' homes (infants at the age of 3 to 8 months)



a

1 M: a) ha?
huh?

Rączaszek-Leonardi et al., 2013



b

2 [b] (0.8 s)

Child‘s hand touches the diaper

Rączaszek-Leonardi et al., 2013



c

3 M: ja du kannst ja schonmal die=
Windel[c)] halten
yes you can start by holding=
the diaper

Mother responds contingently, interprets the child's action, labels the object AND the child's action; affords the child's further actions

Rączaszek-Leonardi et al., 2013

Incrementality

- Partners' contributions **relate** to each other (e.g., Schegloff, 2007)
- Relation allows for meaning and embeddedness

→ Is it the mother who initiates the dialog?

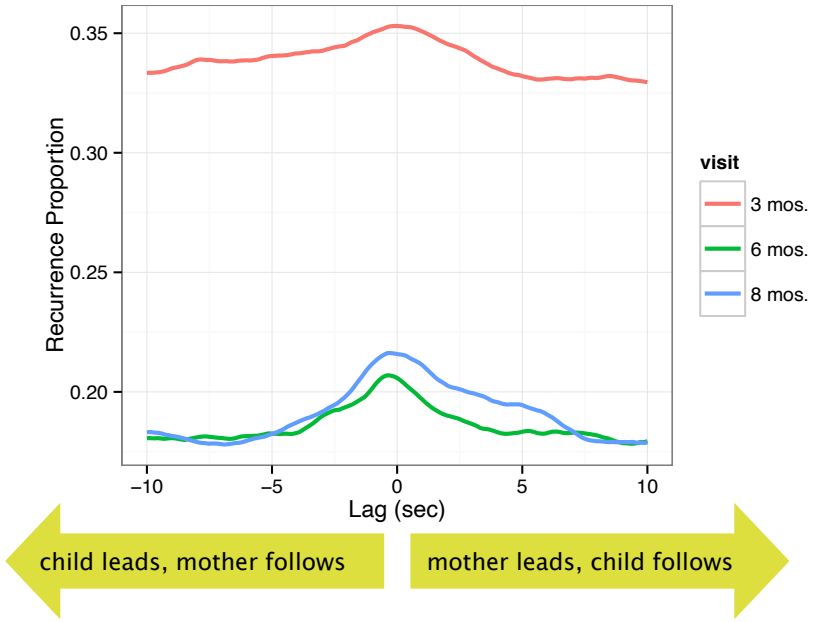
3-month-olds show systematic postural adjustments in interaction (partner-awareness from early on)



Reddy et al., 2013

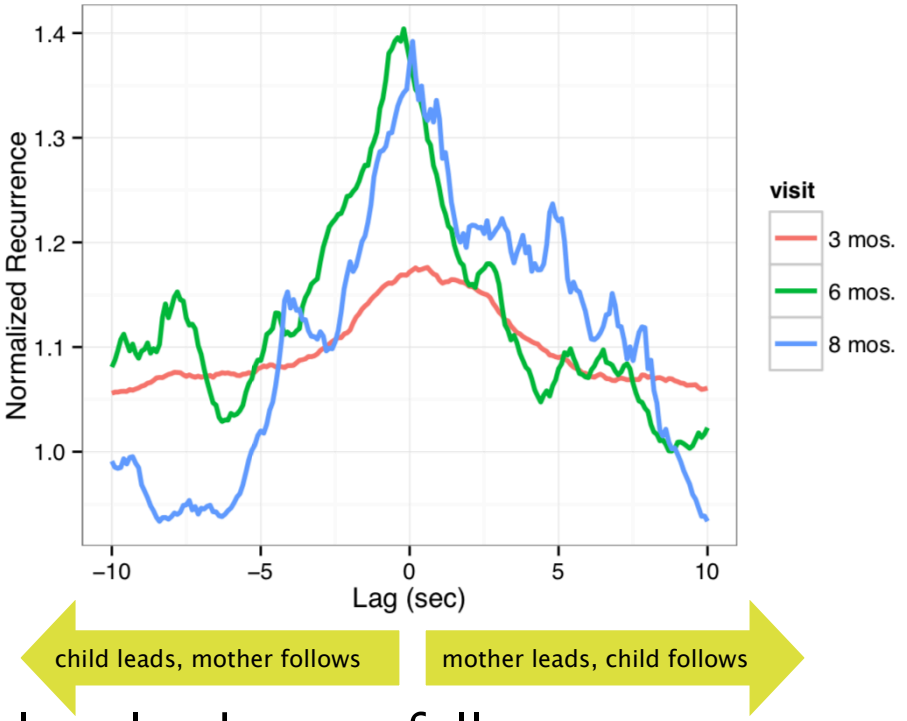
Gaze construction

- Coding of infants and maternal gazes (at the partner, away)
- CRQA-analysis allowing to investigate the dyadic dynamics



Nomikou et al., 2016; Leonardi et al., 2016

Gaze construction



- there is no clear leader nor follower
- less recurrence but more structure at 6 and 8 than 3 months

Nomikou et al., 2016; Leonardi et al., 2016

Incrementality

- Partners' contributions **relate** to each other
- Relation allows for meaning and embeddedness
- Coupling seems at the heart of social interaction

→ Does it allow for adaptation and scaffolding?

video

video



(c) color map (d) intensity map (e) orientation map



(f) flicker map (g) motion map

Learning by interaction

- Cross-sectional study with 47 German learning children at the age of 8–11, 12–23 and 24–30 months and their caregivers
- Longitudinal study with 9 infants
- Applying saliency model (Itti et al., 1998)
- Contribution of the static features to the meaning of action (Nagai & Rohlfing, 2009)

Learning by interaction

start



objects are salient due to motion pauses

middle



face is salient due to speech provided (social information)

end



objects are salient due to pauses and movements‘ inhibition

Learning by interaction

- Caregivers provided more pauses and less round actions to infants
- In addition to attention-getting signals, caregivers provided „checkpoints“ for understanding
- „Motionese“ comprises
  - Attention-getting signals (especially for young children)
  - Action-structuring signals (persist with children's age)

Social partner

- Caregivers are the interpreter of the child's world (Zukow-Goldring, 1994; 2006; Carpenter & Call, 2002)
- With respect to the perceived information, caregivers
  - modify
  - enrich
  - reduce



Rohlfing et al. 2006; Nagai & Rohlfing, 2009; Rohlfing et al., 2022

Incrementality

- Partners' contribution **relate** to each other
- The coupling is at the heart of social interaction
- Relation allows for
  - meaning and embeddedness
  - adaptation
  - scaffolding (in asymmetric interactions)

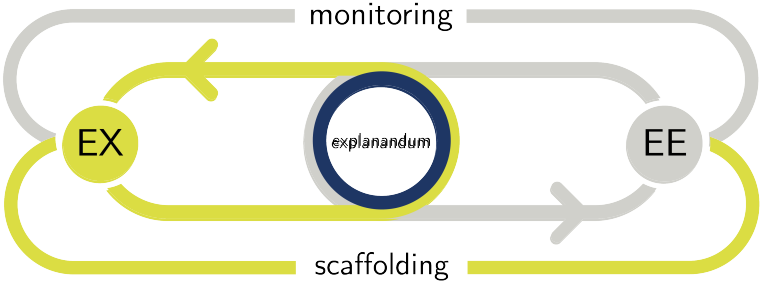


Transfer to XAI

- **Co-construction** as the new paradigm of explaining:

  In essence, co-construction assumes that explainee and explainer actively and cooperatively shape the explanation (Rohlfing et al., 2026)

- Explanation is a cycling between understanding and nonunderstanding (Chi et al., 2001; 2004; Miyake, 1986)
- SNAPE [1]
- SHIFT [2]



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[1] Robrecht et al. (2026) in *Frontiers in Computer Science 8* [2] Groß et al. (2025) in *IEEE ICDL*

Some open questions

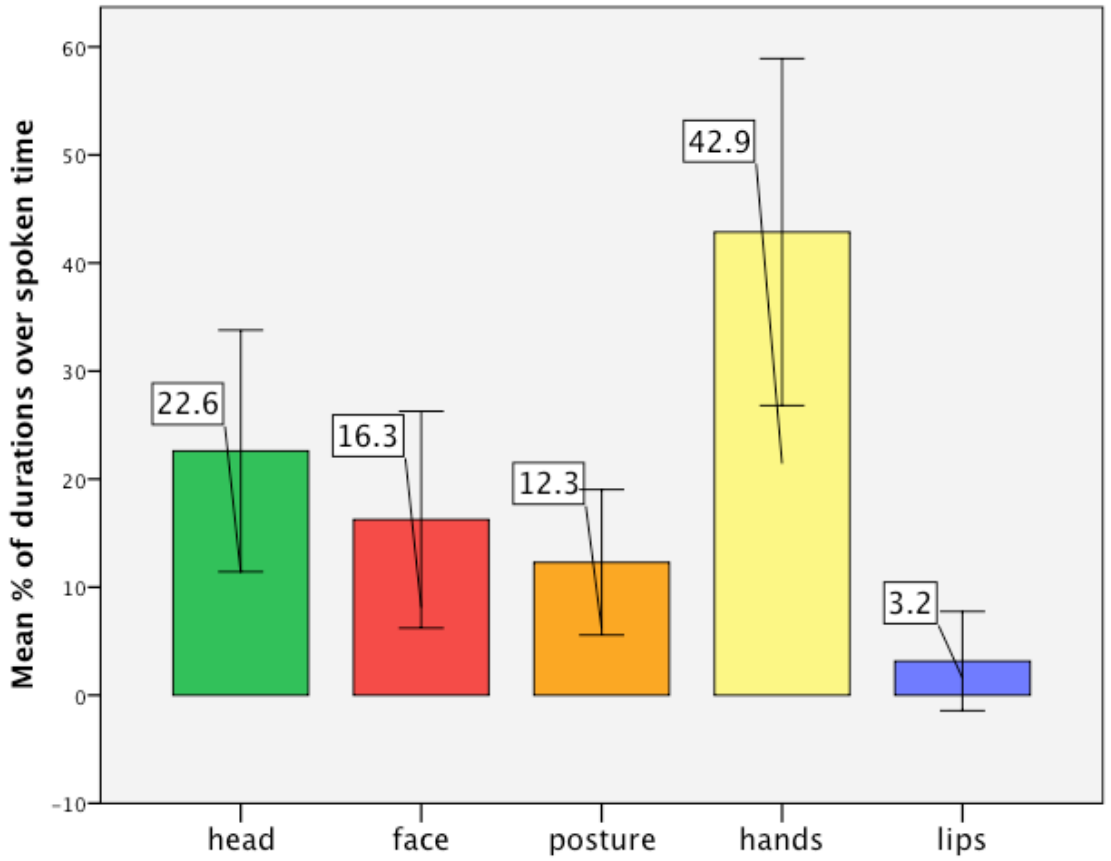
- How to access processuality of interaction?
- How to „fade“ scaffolding?
- How do temperamental differences manifest in interaction?

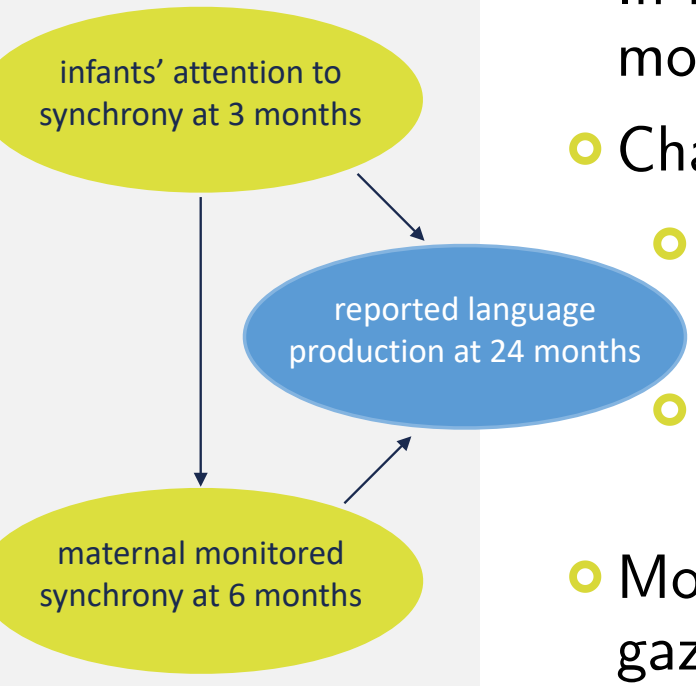
Developmental science innovates XAI research by providing crucial insights into:

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- Incrementality
- **Multimodality**



Iris Nomikou, Joanna Rączaszek-Leonardi, Giuseppe Leonardi, Eyke Hüllermeier

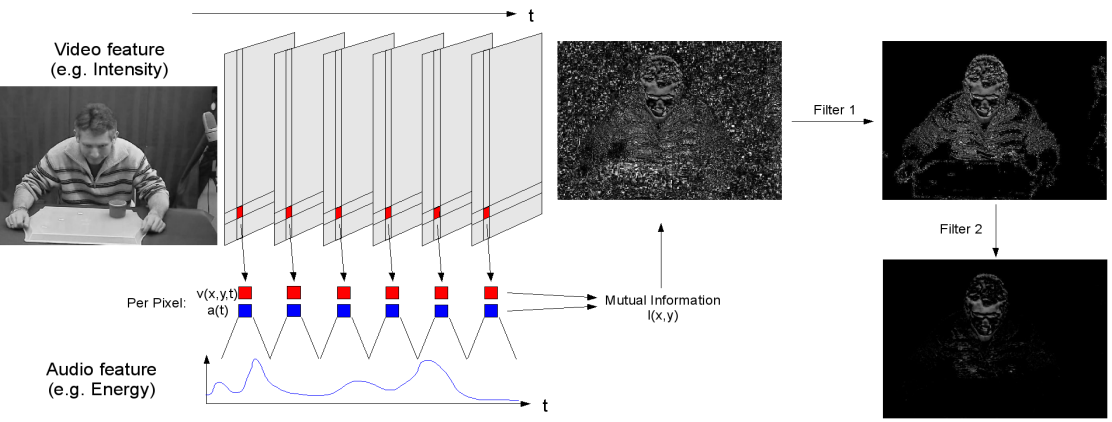

Mean % of durations over spoken time
60
50
40
30
20
10
0
-10
22.6
16.3
12.3
42.9
3.2
head
face
posture
hands
lips



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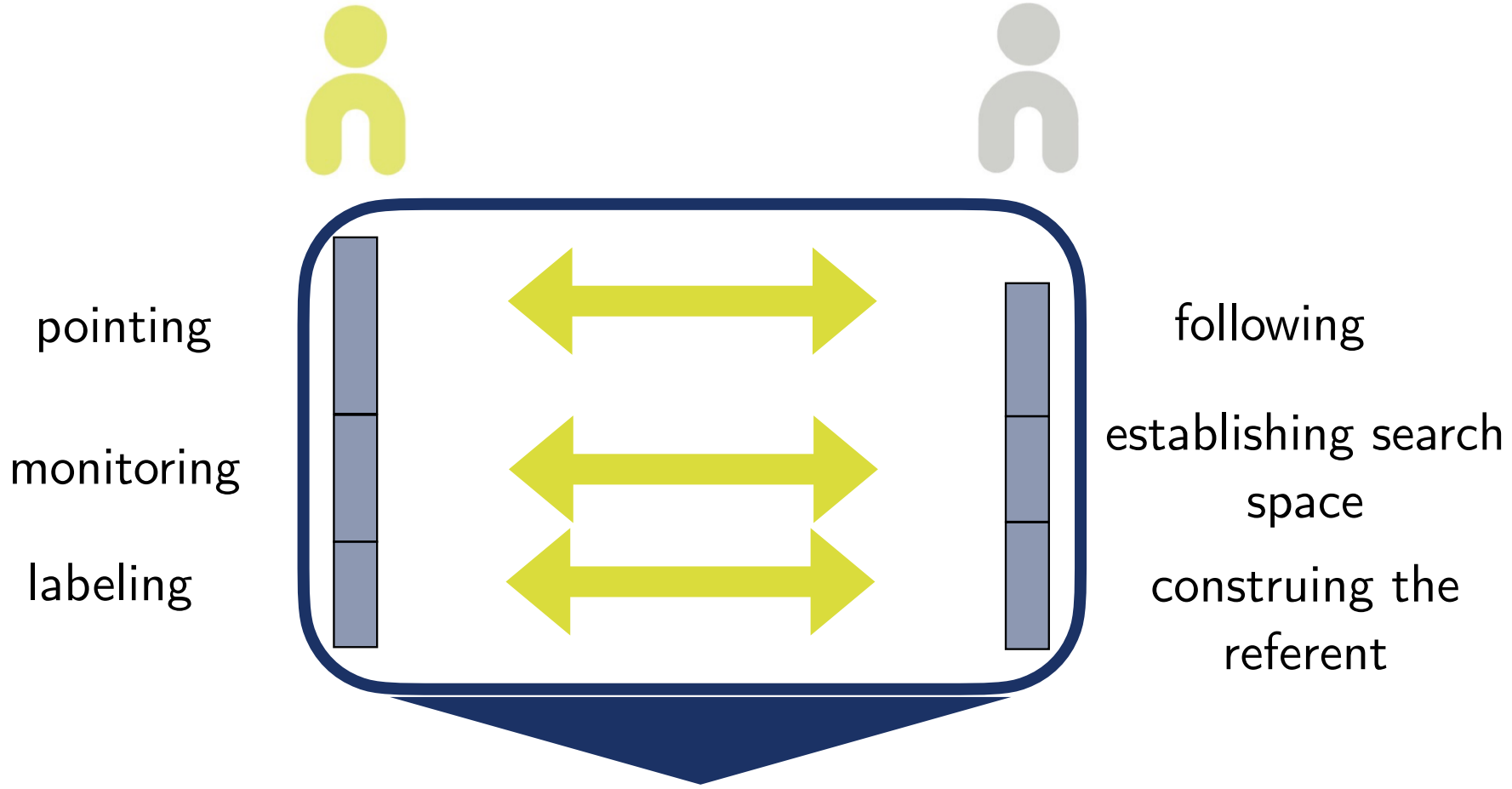
Multimodality: everywhere!

- In mothers, 70 % of vocalized signal is accompanied by bodily movements (head, body, hands) (Nomikou & Rohlfing, 2011)
- Changing roles with the infant's age (Nomikou et al., 2013)
  - At 3 months, infant gazes, mother provides synchronized input
  - At 6 months, mother needs to monitor to provide synchronized input
- More multimodal input that is synchronized with infants' gaze is related to children's later productive vocabulary (Rohlfing & Nomikou, 2014)

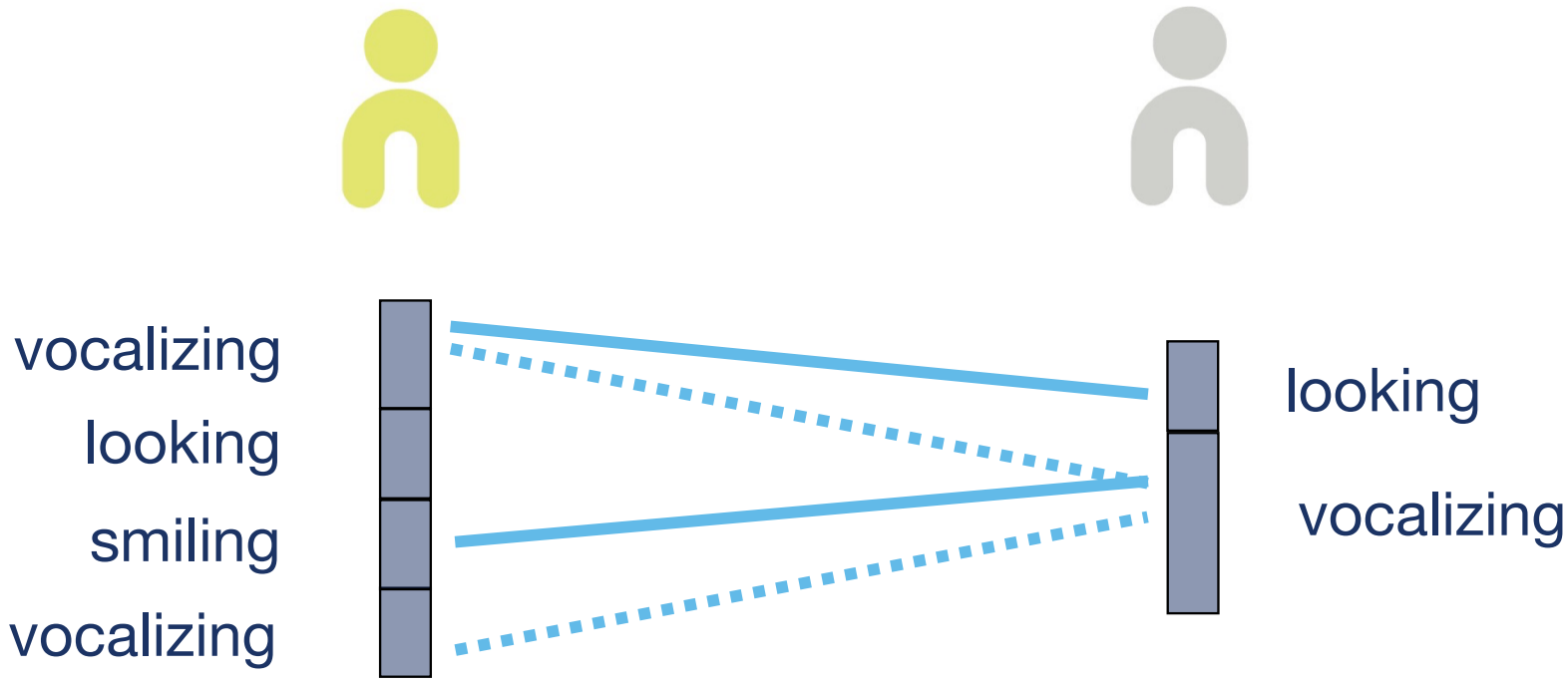


„Educating attention“

- Based on Intersensory Redundancy Hypothesis (Bahrick et al., 2004)
- More synchrony between audio and video features when addressing a child (Rolf et al., 2009)
- Various types of synchrony (Nomikou & Rohlfing, 2011)
- Children categorize objects better when labels are provided (Balaban & Waxman, 1997)

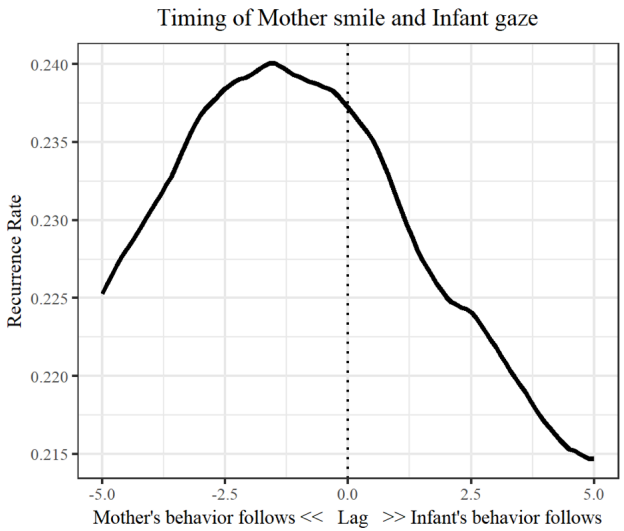


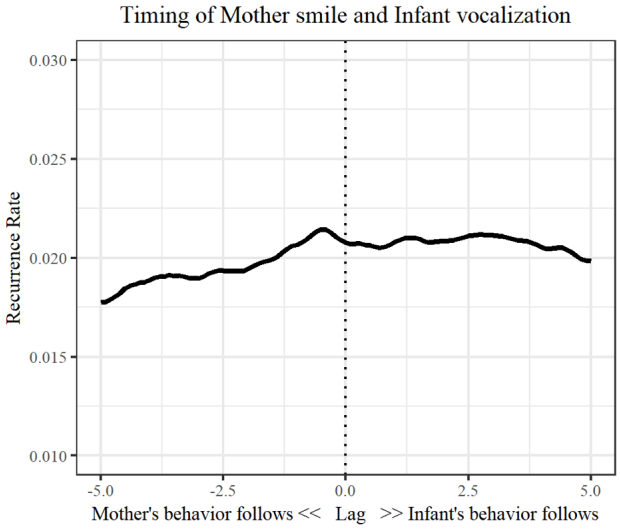
Rohlfing et al., 2016; Heller & Rohlfing, 2017; Rohlfing et al., 2019



Rohlfing et al., 2019, Multimodal turn-taking

CRQA solution

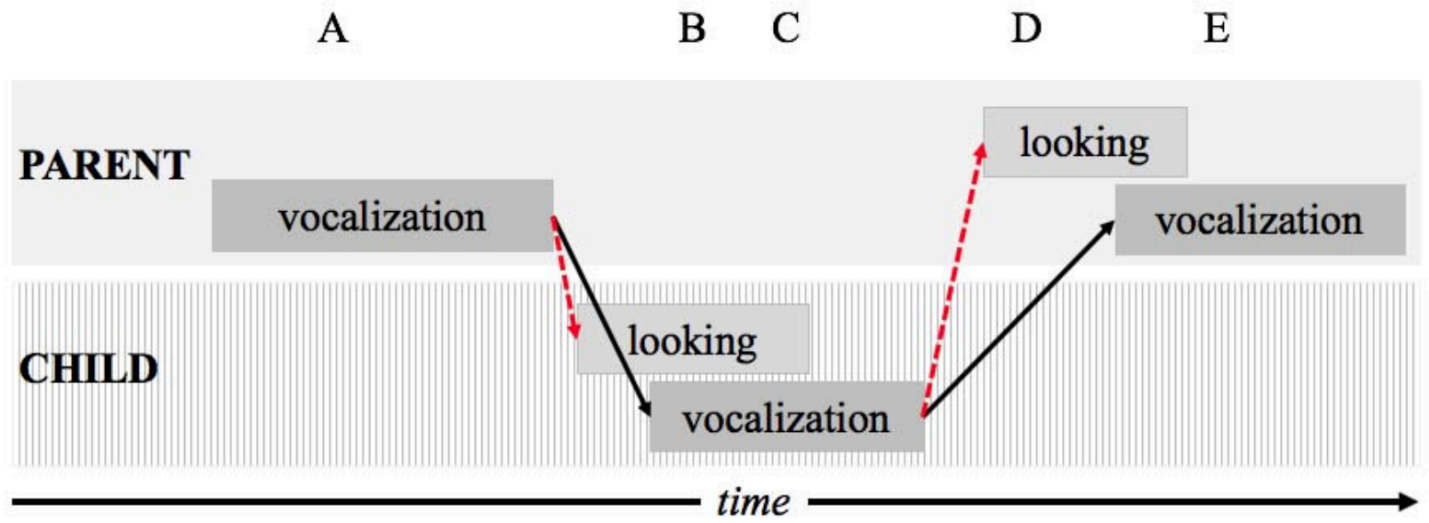




- simplifying behavior in form of binary codes as present or absent
- analysis needs to be run on all combinations

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Pattern mining



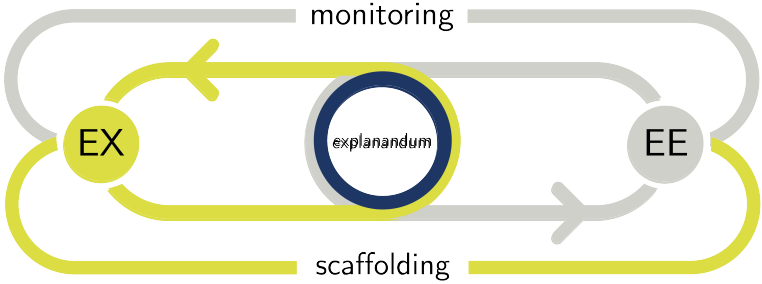
- intrapersonal synchrony: mother smiles — mother speaks
- interpersonal synchrony: mother gazes at infant—infant gazes back—mother speaks

Transfer to XAI

- The process of monitoring uses multimodal signals (Buschmeier et al. at the last CLASP-talk; Lazarov et al., 2026)
- „Ensembles“ (Holler & Levinson, 2019: 641):

  > *signals integrated into a holistic unit based on gestalt-like principles and statistical regularities in the co-occurrence* (cf. Lazarov et al., 2026: 233)

- Scaffolding strategies for a close loop in joint actions (Singh & Rohlfing, 2025; Groß et al., 2023)

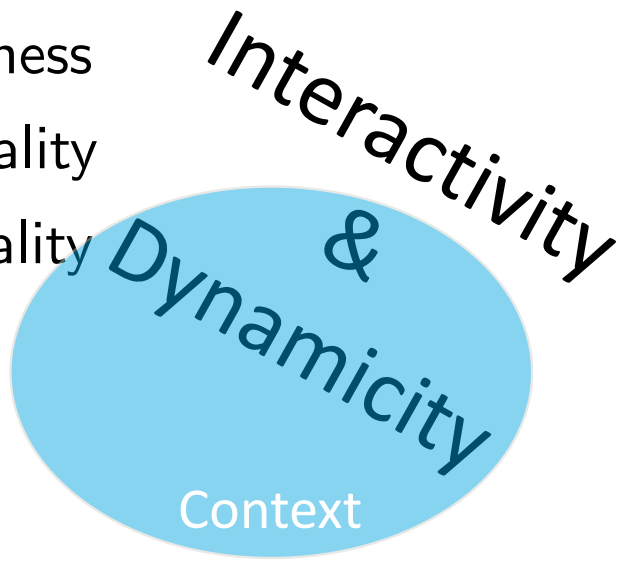


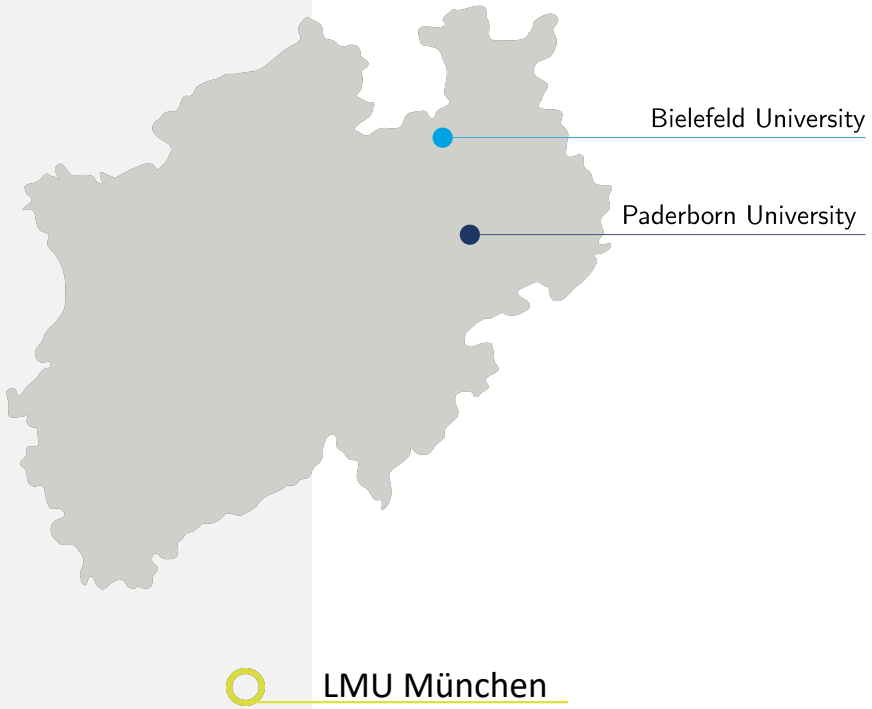
Some open questions

- How can (non)understanding become visible in „ensembles“?
- What are the cross-cultural differences in explanations and „ensembles“?

Developmental science innovates XAI research by providing crucial insights into:

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TRR: A transregional research center

- **Mission:** Relevant explanations for AI-users to exercise their agency and build knowledge
- Coordinated program (DFG) between Paderborn and Bielefeld Universities
- Second period: 1.1.2026 to 30.6.2029
- 20 research projects
- 7 disciplines: computer science, economics, linguistics, media studies, philosophy, psychology, sociology,
- Graduate school (RTG)

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Thank you for your interest!



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